

Alphanumeric Series

Logical Reasoning – Complete Study Guide

Alphanumeric Series questions combine letters (A–Z), numbers (0–9), and sometimes symbols in a sequence. You need to identify a pattern and answer questions such as:

- What comes next?
- Which element is 5th to the left/right?
- How many numbers are immediately preceded by a letter?
- How many letters are between two numbers?
- Which element is exactly midway?
- Which element should replace “?”

1. First, learn the basic codes

Alphabet positions

Letter	Position	Letter	Position
A	1	N	14
B	2	O	15
C	3	P	16
D	4	Q	17
E	5	R	18
F	6	S	19
G	7	T	20
H	8	U	21
I	9	V	22
J	10	W	23
K	11	X	24
L	12	Y	25
M	13	Z	26

Important shortcut: Memorize A=1, E=5, I=9, M=13, Q=17, U=21, Y=25. They are spaced by 4, so they help you quickly calculate nearby positions.

For reverse positions: Z=1, Y=2, X=3, ... A=26

Formula: Reverse position = 27 – normal position

Example: T = 20. Reverse position: $27 - 20 = 7$, so T is 7th from the right.

2. Understand the structure first

Suppose:

A 4 B 7 C 9 D 12 E 15

Separate it mentally:

- Letters → A B C D E
- Numbers → 4 7 9 12 15

Don't try to solve everything simultaneously.

Trick: Separate → Identify pattern → Apply question

3. Main types of Alphanumeric Series

Type 1: Simple alternating pattern

A 2 B 4 C 6 D 8 E 10

- Letters: A, B, C, D, E → increasing by 1
- Numbers: 2, 4, 6, 8, 10 → increasing by 2

If asked: *What is the 7th element?* Count: A, 2, B, 4, C, 6, D. **Answer: D**

4. Trick for Position Questions

A 5 B 7 C 9 D 11 E 13 F 15

Question: *Which element is 4th to the right of B?*

First locate B – it is at position 3. Move 4 places right: 1 → 7, 2 → C, 3 → 9, 4 → D. **Answer: D**

Golden rule: Don't count the starting element. If asked 3rd to the right of X, start counting after X.

5. Trick for “How many elements?”

A 7 B C 9 D 4 E F 8 G

Question: *How many letters are immediately followed by a number?*

Letter	Next element	Result
A	7	Yes
B	C	No
C	9	Yes
D	4	Yes
E	F	No
F	8	Yes
G	(nothing)	No

Answer = 4

Shortcut: Scan only the required relationship (Letter → Number). Don't analyze the entire series.

6. “Immediately preceded by” Trick

4 A 7 B 9 C D 5 E

Question: *How many letters are immediately preceded by a number?*

Pair	Result
4 → A	Yes
7 → B	Yes

Pair	Result
9 → C	Yes
C → D	No
5 → E	Yes

Answer = 4. Remember: *Immediately* = next position only.

7. “Between” Questions

A 4 B 7 C 9 D

Question: *How many elements are between B and D?*

Positions: A(1) 4(2) B(3) 7(4) C(5) 9(6) D(7). Between B and D: 7, C, 9. **Answer = 3**

Formula: Elements between = $|P2 - P1| - 1$. Example: D = 7, B = 3 → $7 - 3 - 1 = 3$

8. Odd/Even Position Trick

A 3 B 5 C 7 D 9 E 11

Position	1	2	3	4	5	6	7	8	9	10
Element	A	3	B	5	C	7	D	9	E	11

Notice: **odd positions → letters, even positions → numbers.** This is a very common pattern.

Shortcut: If the question asks *What is the 15th element?*, don't count all 15. Since odd positions contain letters, the 15th is the 8th letter = H. **Answer: H**

9. Reverse Counting Trick

A 4 B 7 C 9 D 12 E

Question: *What is the 3rd element from the right?* Count from right: E, 12, D. **Answer: D**

Shortcut: If there are N total elements and you need the Kth from right: Position from left = $N - K + 1$. Example: Total = 10, 5th from right → $10 - 5 + 1 = 6$, so find the 6th element from left.

10. Alphabet Pattern Trick

Sometimes letters aren't consecutive.

A 2 D 4 G 6 J 8 M

- Letters: A → D → G → J → M. Positions: 1 → 4 → 7 → 10 → 13. Pattern: +3
- Numbers: 2 → 4 → 6 → 8. Pattern: +2

Next letter after M: $M + 3 = \mathbf{P}$

11. Reverse Alphabet Pattern

Z 2 W 4 T 6 Q 8 N

- Letters: Z → W → T → Q → N. Positions: 26 → 23 → 20 → 17 → 14. Pattern: -3

Next: $N - 3 = \mathbf{K}$

12. Mixed Pattern Trick

A 3 D 6 G 9 J 12 M

- Letters: A → D → G → J → M (+3)
- Numbers: 3 → 6 → 9 → 12 (+3)

Both follow their own pattern.

Golden method for complicated series:

Step 1: Separate letters

Step 2: Separate numbers

Step 3: Check each pattern independently

Step 4: Recombine if necessary

13. Symbol-Based Questions

A @ 5 B # 7 C \$ 9 D % 4 E

The series contains letters, numbers, and symbols. If asked: *Which symbol is immediately preceded by a number?*

Check: 5 B → no; 7 C → no; 9 D → no; 4 E → no. There is no symbol immediately after a number.

Don't get distracted by symbols that aren't relevant to the question.

14. Group Pattern Trick

A1B2C3 D4E5F6 G7H8I9

You can recognize groups: A1B2C3, D4E5F6, G7H8I9. Each group follows Letter → Number, which makes position questions much faster.

15. Most Important Exam Tricks

Trick 1 - Number the positions

For difficult series, immediately write 1 2 3 4 5 6 7 8... and then write the elements below them. This prevents counting mistakes.

Trick 2 - Separate categories

For A 5 D 8 G 11 J 14, analyze Letters → A D G J (+3) and Numbers → 5 8 11 14 (+3) separately.

Trick 3 - “Immediately” means one place

Immediately left/right = exactly one position. Don't count two or three positions.

Trick 4 - “Between” excludes both endpoints

A _ B _ C: between A and C is B only. Answer = 1.

Trick 5 - Reverse questions

Use the formula: Kth from right = N – K + 1 from left. This saves time.

Trick 6 - Convert letters to numbers

For complex questions use A=1, B=2, ..., Z=26. Then look for: +1, +2, +3, -1, -2, -3, odd/even, alternate patterns.

16. How to Prepare for Company-Level Questions

For aptitude/interview preparation, practice in this order:

Level	Practice
Level 1 – Basic	Alphabet positions; number positions; left/right counting; before/after; between; odd/even positions
Level 2 – Intermediate	Alternating letters/numbers; reverse alphabet; missing elements; multiple patterns; symbol combinations; position-based questions
Level 3 – Advanced	Multiple alternating patterns; letter + number transformations; complex position calculations; conditional counting; multiple statements; questions requiring 2–3 operations

17. Fast Solving Strategy for Exams

STEP 1 – Read the entire series once.

STEP 2 – Identify the question type:

If the question is...	Do this
Position?	Count
Between?	Position difference – 1
Immediately?	Check adjacent element
Pattern?	Separate letters/numbers
Reverse?	Use $N - K + 1$

STEP 3 – Don't solve unnecessary information.

A 4 C 8 F 12 I 16

Question: *How many numbers are immediately preceded by a letter?* Check A→4, C→8, F→12, I→16: all yes. **Answer = 4.** You don't need to find the overall pattern.

Quick Cheat Sheet

Question wording	What to do
5th to the right	Move 5 places right
3rd to the left	Move 3 places left
4th from right	$N - 4 + 1$
Between A and B	Difference $- 1$
Immediately after	Check next position
Immediately before	Check previous position
Number preceded by letter	Check Letter $\rightarrow$ Number
Letter followed by number	Check Letter $\rightarrow$ Number
Odd position	1, 3, 5, 7...
Even position	2, 4, 6, 8...
Alphabet pattern	Convert A-Z to 1-26
Reverse alphabet	$Z=26 \dots A=1$ (reverse position = $27 - \text{normal}$)
Complex series	Separate letters / numbers / symbols